

ECON 144 — PRACTICE FINAL EXAM SOLUTION KEY SECTION 1: Time Series Fundamentals 1. Weak stationarity: mean constant, variance constant, autocovariance depends only on lag. 2. ACF: slow decay indicates AR component; cutoff indicates MA. PACF: cutoff indicates AR order. SECTION 2: Trend/Seasonality 3. Linear model $y = \beta_0 + \beta_1 t$; β_1 = average growth per period. 4. Additive seasonality: constant amplitude. Multiplicative: proportional amplitude. SECTION 3: ARIMA Modeling 5. AR(p): PACF cuts off at p. MA(q): ACF cuts off at q. 6. Differencing removes stochastic trends; unit root implies nonstationary. SECTION 4: ARCH/GARCH 7. Persistence = $\alpha + \beta$. 8. Forecast: $\sigma^2_{\{t+1\}} = \omega + \alpha \varepsilon^2_t + \beta \sigma^2_t$. SECTION 5: Forecast Evaluation 9. DM test: $DM = \text{mean}(d) / NW_se$; reject if $|DM| > 1.96 \rightarrow$ different accuracy. 10. MZ: $y = \alpha + \beta f$. $H_0: \alpha = 0, \beta = 1$. Bias if $\alpha \neq 0$; inefficiency if $\beta \neq 1$. SECTION 6: Cointegration & ECM 11. EG step 1: regress y on x; step 2: ADF test residuals. Reject $H_0 \rightarrow$ cointegration. 12. ECM: $\Delta y = \alpha + \beta \Delta x + \lambda u_{\{t-1\}}$. λ = speed of adjustment. SECTION 7: Kalman Filter 13. Q = variance of state noise; R = variance of observation noise. Kalman gain increases with Q and decreases with R. 14. Time-varying β smoothness depends on Q (small Q = very smooth). SECTION 8: MCMC & Markov Chains 15. Gibbs sampling: alternately sample from conditional distributions. 16. Stationary distribution for 2-state chain: solve $\pi = \pi P$ with $\pi_1 + \pi_2 = 1$.